

To solve the problem, we need to determine the maximum index N across all multisets S of positive integers less than 20, then find the smallest maximum element M for multisets achieving this N , and finally the smallest size of such a multiset.

Step 1: Determine the maximum index N .

The index of a multiset S is the largest n such that for every i from 1 to n , S can be partitioned into multisets A_i and B_i where $A_i = i + B_i$.

- For $N = 3$, consider the multiset $S = \{1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6\}$. This set can be partitioned for $i = 1, 2, 3$:
 - $i = 1$: Partition into $B_1 = \{1, 1, 3, 3, 5, 5\}$ and $A_1 = \{2, 2, 4, 4, 6, 6\}$. Each element in A_1 is $1 +$ an element in B_1 .
 - $i = 2$: Partition into $B_2 = \{1, 1, 2, 2, 3, 3\}$ and $A_2 = \{3, 3, 4, 4, 5, 5\}$. Each element in A_2 is $2 +$ an element in B_2 .
 - $i = 3$: Partition into $B_3 = \{1, 1, 2, 2, 3, 3\}$ and $A_3 = \{4, 4, 5, 5, 6, 6\}$. Each element in A_3 is $3 +$ an element in B_3 .
- For $N = 4$, no such multiset exists. For example, consider $S = \{1, 2, \dots, 16\}$. Partitioning for $i = 4$ requires elements like 13 or 17, which are outside the allowed range or not present. Thus, $N = 3$ is the maximum.

Step 2: Determine the smallest maximum element M .

To minimize M , we construct a multiset achieving $N = 3$ with the smallest possible maximum element. The multiset $S = \{1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6\}$ has $M = 6$. Any smaller M (e.g., $M = 5$) fails to satisfy the partition for $i = 3$, as elements like 6 would be required but missing.

Step 3: Determine the smallest size of S .

The multiset $S = \{1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6\}$ has 12 elements. Reducing the size below 12 (to 10 or fewer) while maintaining $N = 3$ and $M = 6$ is impossible. For example:

- A size-10 multiset like $\{1, 1, 2, 2, 3, 3, 4, 4, 5, 6\}$ lacks sufficient elements to partition for $i = 3$.

Thus, the smallest possible size is 12.

Final Answer: $\boxed{12}$